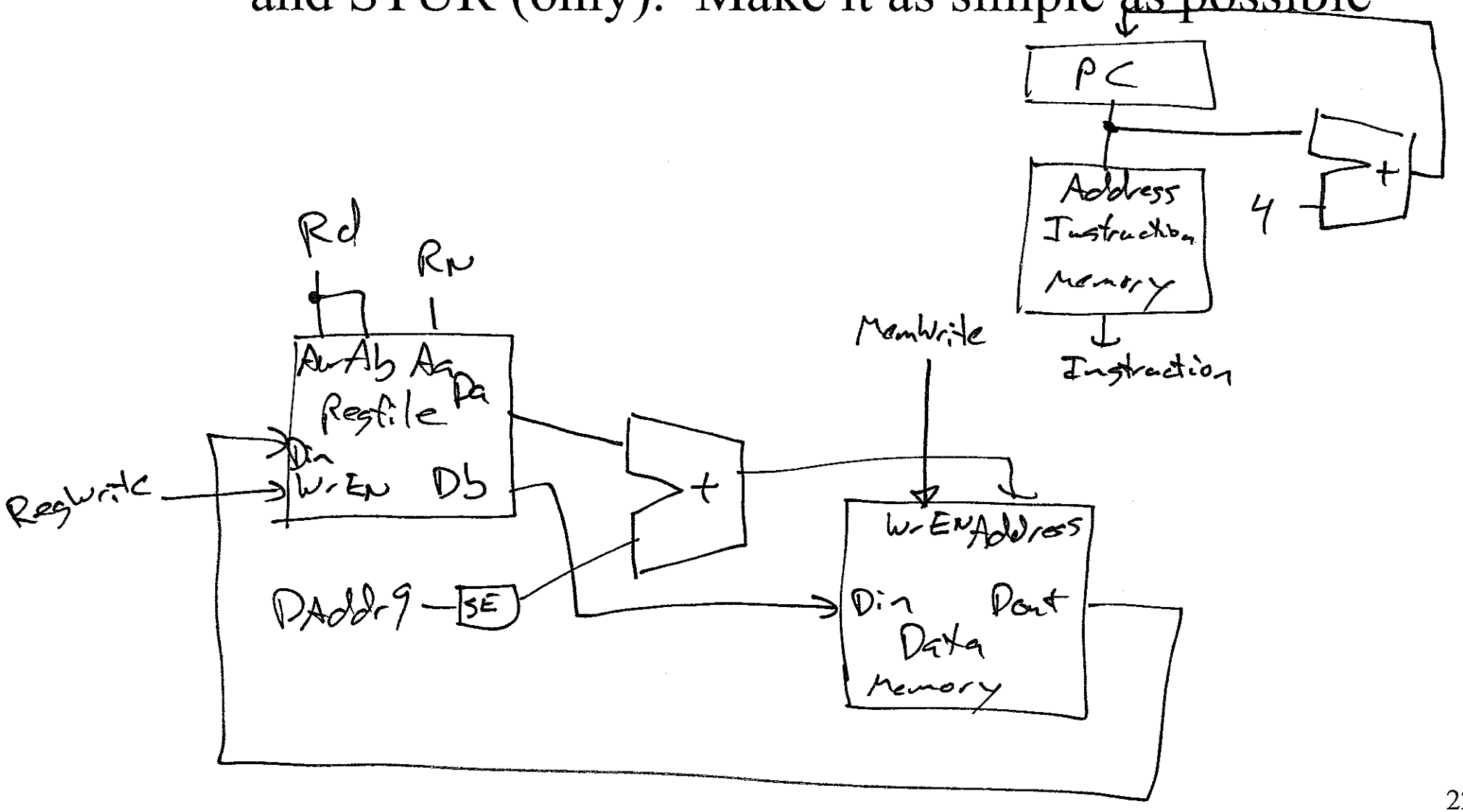
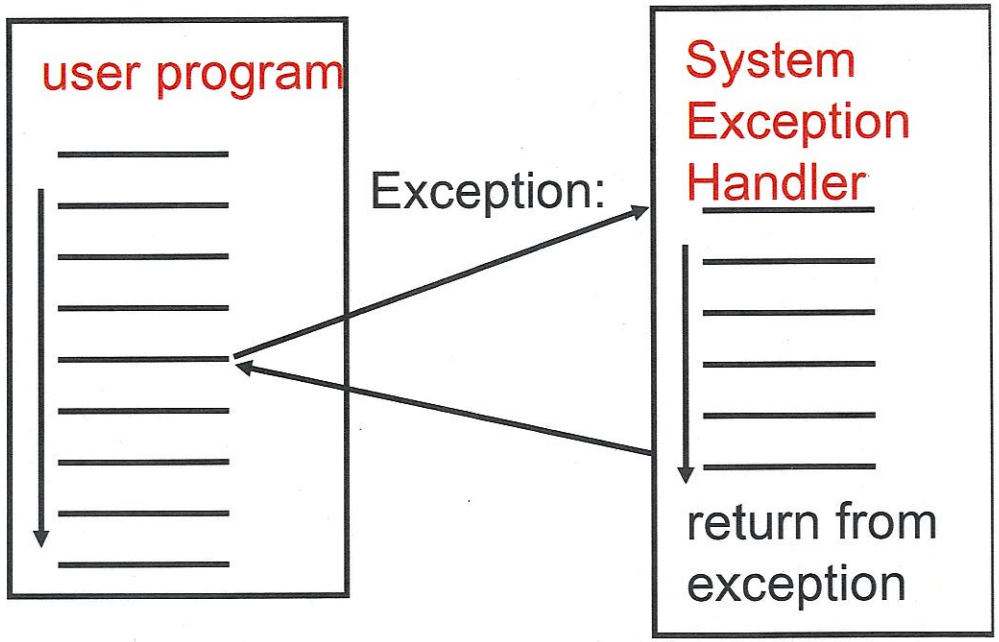
Review Problem 22

- Develop a single-cycle CPU that can do LDUR and STUR (only). Make it as simple as possible



Advanced: Exceptions

Exception = unusual event in processor
- Arithmetic overflow, divide by zero, …
- Call an undefined instruction
- Hardware failure
- I/O device request (called an “interrupt”)



Approaches
- Make software test for exceptional events when they may occur (“polling”)
- Have hardware detect these events & react:
  - Save state (Exception Program Counter, protect the GPRs, note cause)
  - Call Operating System
    - If (undef_instr) PC = C0000000
    - If (overflow) PC = C0000020
    - If (I/O) PC = C0000040
    - …

Performance of Single-Cycle Machine

CPI?

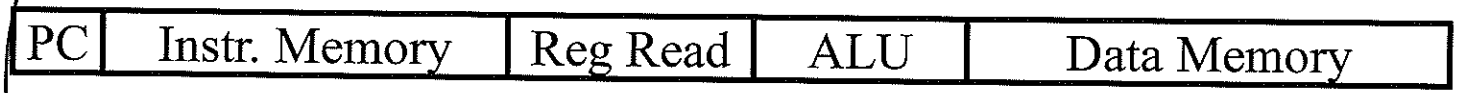
ADD, SUB



LDUR



STUR



CBZ



B

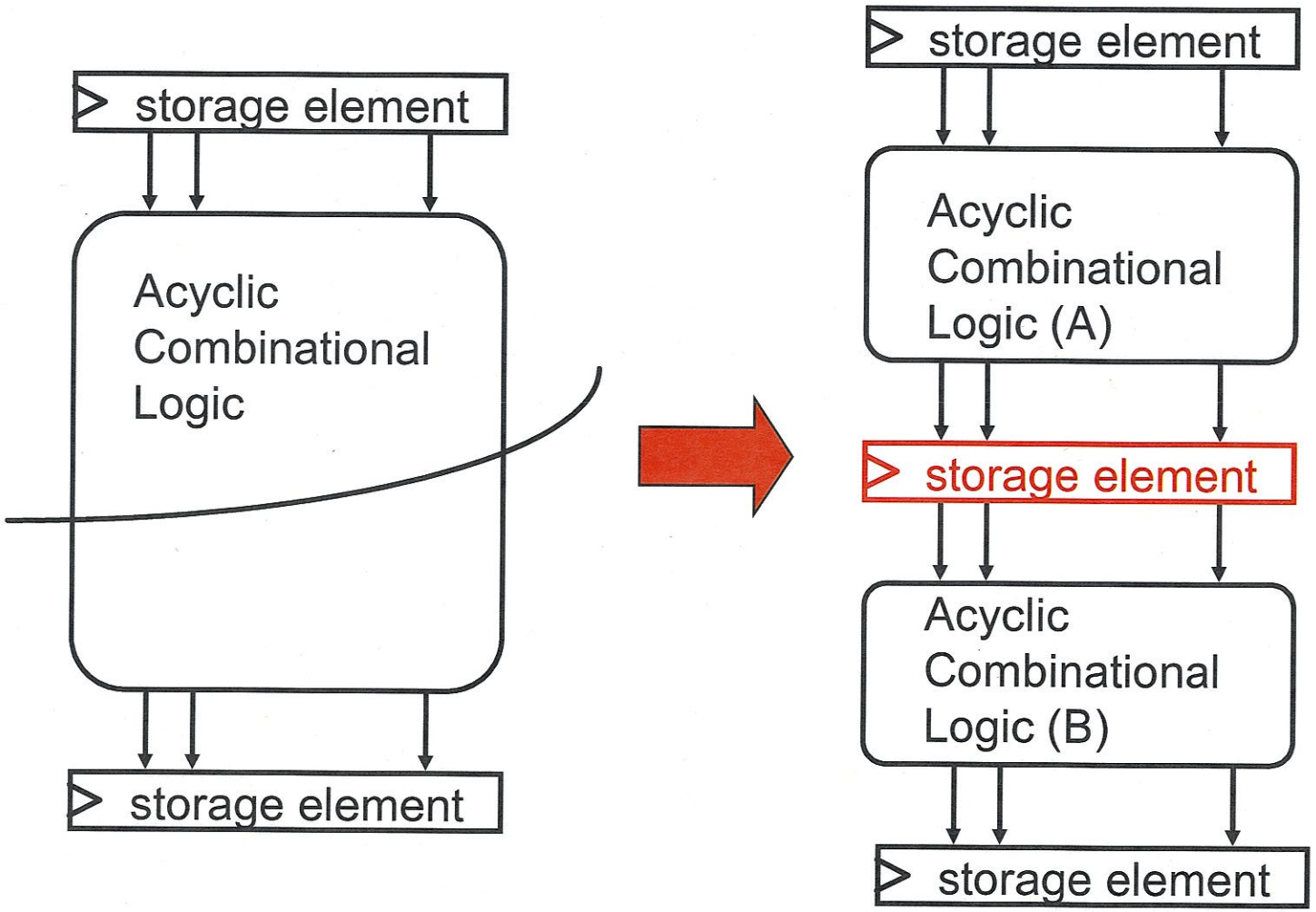


clock period

Reducing Cycle Time

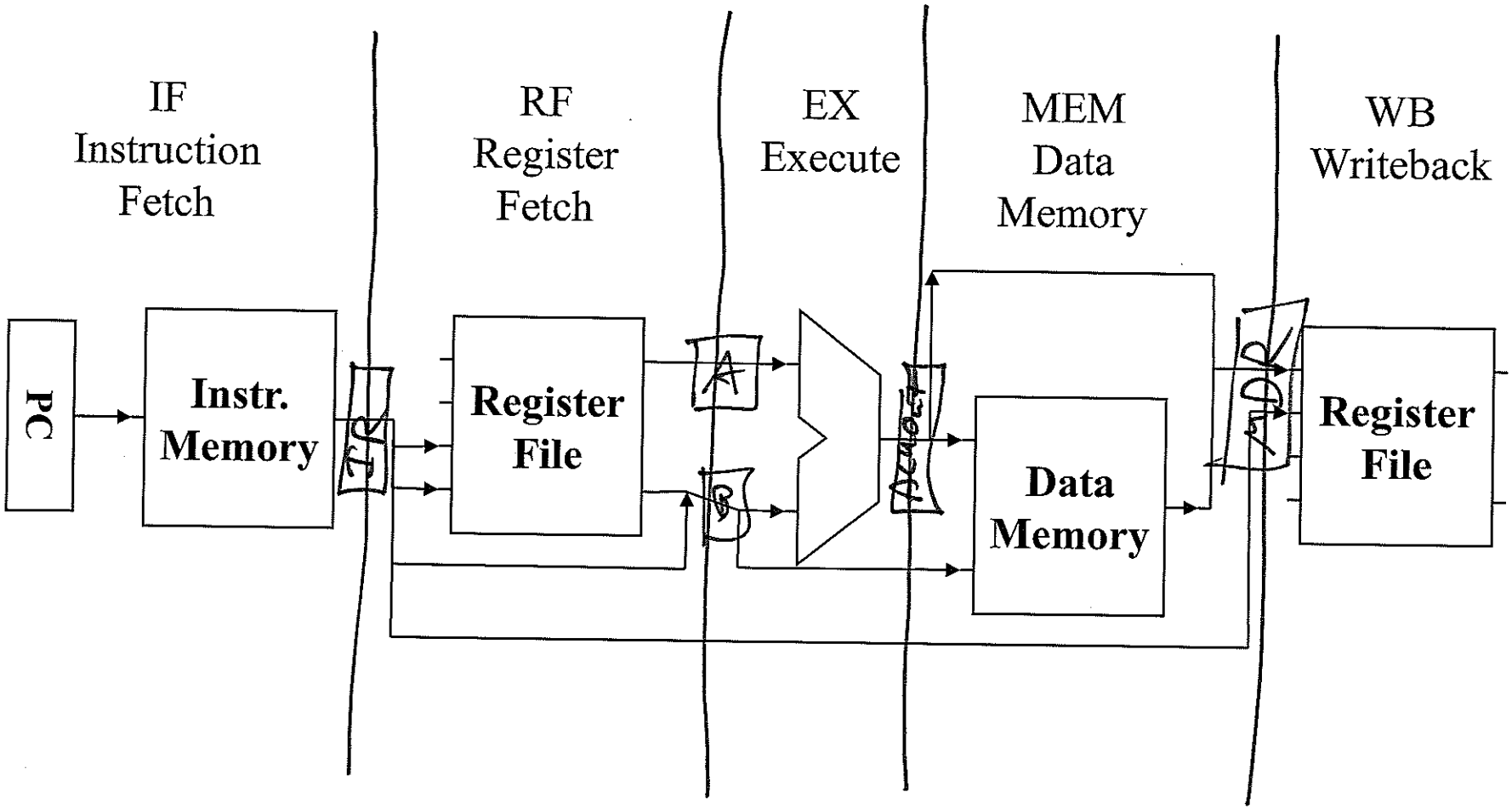
Cut combinational dependency graph and insert register / latch

Do same work in two fast cycles, rather than one slow one



Pipelined Processor Overview

Divide datapath into multiple stages



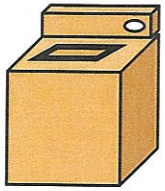
Pipelining

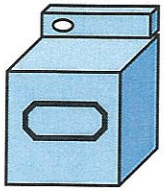
Readings: 4.5-4.8

Example: Doing the laundry

Ann, Brian, Cathy, & Dave 

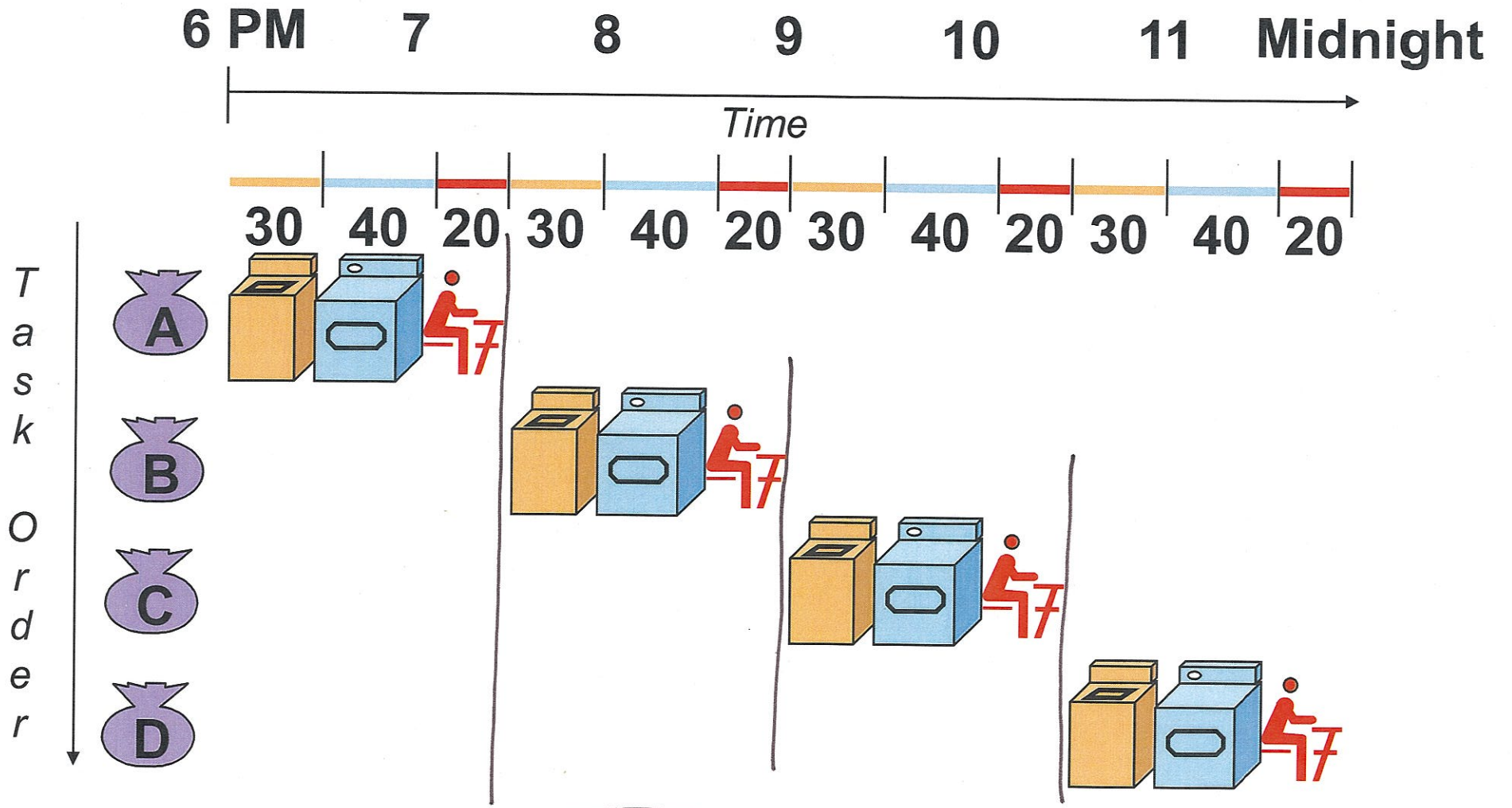
each have one load of clothes to wash, dry, and fold

Washer takes 30 minutes 

Dryer takes 40 minutes 

"Folder" takes 20 minutes 

Sequential Laundry



Sequential laundry takes 6 hours for 4 loads

If they learned pipelining, how long would laundry take?

Pipelined Laundry: Start work ASAP

